

BTEB LEVEL-3: WEB DESIGN (EASY REVISION GUIDE)

Super Simple Words | 1-Line Answers | Easy Examples | Full A-to-Z Coverage

[PDF: Question & Viva.pdf] & [PDF: NSDA Assessment.pdf] — মূল ফাইনাল বোর্ড পরীক্ষা

২০টি MCQ + ১০টি Short Questions + ১০টি Viva Questions

অংশ ক: ২০টি MCQ (খুব সহজ ভাষায়)

1. Why is it important to consider all parties in negotiations?

a) To ensure fairness | b) All of the above | c) To achieve a mutual agreement | d) To avoid conflict

👉 উত্তর: b) All of the above

সহজ বাংলা: সবার কথা ভাবলে চুক্তি ভালো হয় এবং কোনো ঝগড়া হয় না।

2. Common health risk in IT workplaces?

a) Hearing loss | b) Risk of drowning | c) Extreme cold | d) Repetitive Strain Injury (RSI)

👉 উত্তর: d) Repetitive Strain Injury (RSI)

সহজ বাংলা: অনেকক্ষণ কীবোর্ড ও মাউস চালালে হাতের আঙুল ও কবজিতে ব্যথা হয় (RSI)।

3. How is GitHub used in web development?

a) Designing graphics | b) Writing content | c) Creating animations | d) Hosting and collaborating on code

👉 উত্তর: d) Hosting and collaborating on code

সহজ বাংলা: কোড সেভ করে রাখা এবং অন্যদের সাথে মিলে কাজ করার জন্য গিটহাব ব্যবহার করা হয়।

4. What is the default port for HTTP?

a) 22 | b) 80 | c) 443 | d) 3306

👉 উত্তর: b) 80 (নোট: HTTPS এর পোর্ট 443)

5. Web-safe colors are:

a) Colors that display consistently across browsers and devices | b) Colors for print | c) Random colors | d) Invisible

👉 উত্তর: a) Colors that display consistently across browsers and devices

সহজ বাংলা: যে ২১৬টি কালার সব মোবাইল ও কম্পিউটারে দেখতে একই রকম লাগে।

6. Offline tools to manage color palettes:

a) Terminal | b) Adobe Illustrator, Photoshop, or Sketch | c) Browser cache | d) HTML editor

👉 উত্তর: b) Adobe Illustrator, Photoshop, or Sketch

7. Which color is commonly used for warning signs or alerts?

a) Green | b) Blue | c) Red | d) Purple

👉 উত্তর: c) Red (লাল রঙ বিপদ বা অ্যালার্ট বোঝায়)

8. Which accessibility principle focuses on readable color contrast?

a) Robust | b) Understandable | c) Perceivable | d) Operable

👉 উত্তর: c) Perceivable (সহজে চোখে দেখা যায় এমন)

9. Tag used to define a table header?

a) <tr> | b) <td> | c) <th> | d) <table>

👉 উত্তর: c) <th> (Table Header)

10. Link to another part of the SAME page?

a) Link | b) | c) | d)

👉 উত্তর: a) Link (হ্যাস # চিহ্ন দিয়ে id লিঙ্ক করা হয়)

11. What does <head> section contain?

a) Main visible content | b) Stylesheets, scripts, and metadata | c) Only images | d) Footer

👉 উত্তর: b) Stylesheets, scripts, and metadata

12. Center a background image in CSS?

a) background-align: center; | b) background: center-image; | c) background-position: center; | d) image-align: middle;

👉 উত্তর: c) background-position: center;

13. Make a div take full screen height?

a) height: 100vh; | b) height: 100%; | c) height: full; | d) height: 100vw;

👉 উত্তর: a) height: 100vh; (vh = viewport height)

14. Link external CSS file in HTML?

a) <style src="style.css"> | b) <link rel="stylesheet" href="style.css"> | c) <script href="style.css"> | d) <css>

👉 উত্তর: b) <link rel="stylesheet" href="style.css">

15. Software used for designing web layouts?

a) MS Word | b) VLC | c) Figma | d) Chrome

👉 উত্তর: c) Figma

16. Scalable file format for web icons?

a) JPG | b) PNG | c) SVG | d) BMP

👉 উত্তর: c) SVG (জুম করলেও ফাটে না)

17. Make a website work on mobile and desktop?

a) Fixed width | b) Responsive design with Flexbox or Grid | c) Desktop only | d) Disable mobile

👉 উত্তর: b) Responsive design with Flexbox or Grid

18. Output of: `let x = 10; if (x > 5) { console.log("Yes"); } else { console.log("No"); }`

a) No | b) undefined | c) Error | d) Yes

👉 উত্তর: d) Yes (কারণ ১০ পাঁচের চেয়ে বড়)

19. Difference between `==` and `===`?

a) No difference | b) `==` checks value only, `===` checks value and data type | c) `===` checks value only | d) `==` is assignment

👉 উত্তর: b) `==` checks value only, `===` checks value and data type

20. What is an open-ended question?

a) Answered with Yes/No | b) Allows client to explain in detail | c) MCQ | d) No answer

👉 উত্তর: b) Allows client to explain in detail (যেমন: "আপনার ওয়েবসাইটে কী কী ফিচার চান?")

অংশ থ: ১০টি শর্ট প্রশ্ন (সহজ ১ লাইনের উত্তর ও কোড)

21. Mention one electrical safety hazard in the workplace.

👉 Ans: Overloading a power socket with too many plugs.

সহজ বাংলা: একটা মাল্টিপ্লাগে অনেক বেশি কম্পিউটার লাগানো।

22. What is a media query in CSS?

👉 Ans: CSS code used to make web pages look good on all screen sizes (mobile, tablet, PC).

Ex: `@media (max-width: 600px) { body { font-size: 14px; } }`

সহজ বাংলা: স্ক্রিনের সাইজ অনুযায়ী ডিজাইন বদলানোর সিএসএস কোড।

23. What is modular design in UI?

👉 Ans: Breaking a web page layout into small reusable components.

Ex: Making one reusable `.card` component used on many pages.

সহজ বাংলা: পুরো ওয়েবসাইটকে ছোট ছোট ব্লকে (Navbar, Card, Footer) ভাগ করে তৈরি করা।

24. What is mobile-first design?

👉 Ans: Designing the website for mobile screens first, then expanding for desktop.

সহজ বাংলা: আগে মোবাইলের স্ক্রিন অনুযায়ী ডিজাইন করা, তারপর বড় স্ক্রিনের জন্য সাজানো।

25. Semantic vs Non-semantic HTML?

👉 Ans: Semantic tags have clear meaning (`<header>`, `<footer>`). Non-semantic tags have no meaning (`<div>`, `<span>`).

Ex: Semantic: `<header>` | Non-semantic: `<div class="header">`

26. How do you select elements by class in CSS?

👉 Ans: Using a dot (`.`) followed by class name.

Ex: `.my-btn { color: red; }`

27. Highlight the first letter of a paragraph in CSS.

👉 Ans: Using `::first-letter` selector.

```
Ex: p::first-letter { font-size: 24px; color: red; }
```

28. Simple HTML Form code with class name:

```
<form class="my-form"> <input type="text" name="user" placeholder="Name"> <button  
type="submit">Submit</button> </form>
```

29. What is visual hierarchy?

👉 Ans: Arranging page items by importance so users see the most important thing first.

Ex: Heading is large (h1 32px), normal text is small (p 14px).

সহজ বাংলা: গুরুত্বপূর্ণ লেখা বড় আর কম গুরুত্বপূর্ণ লেখা ছোট রাখা।

30. What is a loop in JavaScript?

👉 Ans: Code that runs again and again until a condition stops it.

```
Ex: for (let i = 1; i <= 5; i++) { console.log(i); }
```

1. How can you present a PowerPoint slideshow?

👉 Ans: Press F5 to start from beginning, or Shift + F5 to start from current slide.

2. What is a formula in Excel?

👉 Ans: An expression used for calculation that always starts with an equals sign (=).

Ex: =SUM(A1:A10) calculates total sum.

3. How does jQuery handle events?

👉 Ans: jQuery uses easy methods like .click() and .hover().

Ex: \$("#btn").click(function() { alert("Clicked!"); });

4. What are the advantages of Bootstrap?

👉 Ans: 1) Responsive design, 2) Ready-made components, 3) 12-column grid, 4) Saves development time.

5. Examples of popular CMS platforms?

👉 Ans: WordPress, Joomla, Drupal.

6. Three names of JavaScript data types?

👉 Ans: String, Number, Boolean.

7. Loop used to iterate over Object properties?

👉 Ans: The for...in loop.

Ex: for (let key in person) { console.log(key, person[key]); }

8. HTML tag used for main page content?

👉 Ans: The <main> tag.

9. How to target an element by ID in CSS?

👉 Ans: Using hash (#) symbol.

Ex: #header { background: blue; }

10. CSS property used to change text color?

👉 Ans: The color property.

Ex: p { color: red; }

1. How to ensure accessibility in JavaScript?

👉 Ans: Use semantic HTML, ARIA attributes, and keyboard navigation support.

সহজ বাংলা: কীবোর্ড দিয়ে যাতে পুরো ওয়েবসাইট চালানো যায় এবং অন্ধ মানুষ স্ক্রিন রিডারে পড়তে পারে।

2. Image format supporting transparency?

👉 Ans: PNG (Portable Network Graphics) and SVG.

3. Difference between == and ===?

👉 Ans: == checks only value. === checks value AND data type.

Ex: 5 == "5" is true | 5 === "5" is false

4. What are logical operators?

👉 Ans: Operators used to check multiple conditions: && (AND), || (OR), ! (NOT).

5. How to use break statement?

👉 Ans: In loops: stops and exits loop immediately. In switch: stops next cases from running.

Ex: for (let i=1; i<=5; i++) { if (i===3) break; console.log(i); }

6. How to display output in JavaScript?

👉 Ans: console.log(), alert(), document.write(), innerHTML.

7. Comments in JavaScript?

👉 Ans: Single-line: // comment | Multi-line: /* comment */

8. Basic data types in JavaScript?

👉 Ans: String, Number, Boolean, Undefined, Null, Object.

9. Check data type of variable?

👉 Ans: Using typeof operator.

Ex: typeof "hello" is "string" | typeof 10 is "number"

10. Arithmetic operators in JavaScript?

👉 Ans: + (add), - (subtract), * (multiply), / (divide), % (remainder).

11. Keywords to declare a variable?

👉 Ans: var, let, const.

12. Strict equality operator?

👉 Ans: ===

13. Function to print to console?

👉 Ans: `console.log()`

14. Convert string to integer?

👉 Ans: `parseInt()`

Ex: `parseInt("120px")` returns 120.

15. Tailwind CSS vs Bootstrap?

👉 Ans: Tailwind gives small utility classes for custom styling. Bootstrap gives ready-made components.

সহজ বাংলা: Tailwind-এ ছোট ছোট ক্লাস দিয়ে নিজের মতো ডিজাইন করা যায়। Bootstrap-এ তৈরি জিনিস থাকে।

16. innerText vs innerHTML?

👉 Ans: `innerText` gets only visible plain text. `innerHTML` gets text along with HTML tags.

17. Static vs Dynamic vs Interactive website?

👉 Ans: Static = Fixed content (HTML). Dynamic = Data changes from database. Interactive = Responds to clicks, forms, animations.

18. Why do you prefer VS Code?

👉 Ans: Fast, lightweight, free extensions (Live Server), and built-in terminal.

19. Purpose of FTP in web development?

👉 Ans: To upload and download website files to and from a web hosting server.

20. Simple HTML Login Form code:

```
<form> <input type="text" placeholder="Username"><br> <input type="password" placeholder="Password">
<br> <button type="submit">Login</button> </form>
```

21. When should you use SVG images?

👉 Ans: For logos and icons because SVG never loses quality when made bigger or smaller.

22. What is an ID selector in CSS?

👉 Ans: Targets one specific element using `#id`.

Ex: `#menu { background: black; }`

23. How do you create functions in JavaScript?

Ex: `function greet() { console.log("Hello"); }` Or Arrow Function: `const greet = () => { console.log("Hello"); }`

24. Ordered list in HTML?

Ex: `<ol><li>HTML</li><li>CSS</li></ol>`

25. Array vs Object in JavaScript?

👉 Ans: Array stores a list of values: `[1, 2, 3]`. Object stores key-value pairs: `{name: "Ali", age: 20}`.

26. Three components of Client-Server setup?

👉 Ans: 1) Client (Browser), 2) Server (Computer hosting files), 3) Network (Internet).

27. Why follow Web Standards?

👉 Ans: So that the website looks and works the same on all browsers (Chrome, Firefox, Safari).

28. Role of database in a CMS?

👉 Ans: To store posts, pages, user passwords, and settings safely.

29. Responsive navbar in Bootstrap?

👉 Ans: Uses classes `.navbar`, `.navbar-expand`, and `.navbar-collapse`.

1. What is CSS? How many types?

👉 Ans: CSS styles HTML. 3 Types: 1) Inline (style=""), 2) Internal (<style> in <head>), 3) External (<link href="style.css">).

2. Difference between JavaScript, Bootstrap, and jQuery?

Tool	What it is	Main job	Easy meaning
JavaScript	Programming Language	Adds logic & action	🧠 Brain
Bootstrap	CSS Framework	Design & responsive grid	👁️ Looks
jQuery	JavaScript Library	Shortens JS code	⚡ Shortcut

3. Margin vs Padding?

👉 Ans: Margin = Space OUTSIDE border. Padding = Space INSIDE border.

সহজ বাংলা: মার্জিন উপাদানটির বাইরে জায়গা নেয়, প্যাডিং ভেতরের কন্টেন্টের চারপাশে জায়গা নেয়।

4. Tag vs Element vs Attribute?

👉 Ans: Tag = <p> | Element = <p>Text</p> | Attribute = class="box" inside tag.

5. What is an IDE?

👉 Ans: Software used to write, run, and debug code (e.g., VS Code).

6. For vs While vs Do-While loop?

👉 Ans: for = when loop count is known. while = checks condition before running. do-while = runs at least once before checking.

7. Universal Selector in CSS?

👉 Ans: Represented by asterisk (*). Selects all elements on page.

Ex: * { margin: 0; padding: 0; box-sizing: border-box; }

8. Rowspan in HTML?

👉 Ans: Merges multiple rows into a single table cell.

Ex: <td rowspan="2">Spans 2 rows</td>

9. Video syntax in HTML?

<video controls width="300"> <source src="movie.mp4" type="video/mp4"> </video>

10. Benefits of Headless CMS?

👉 Ans: Faster website speed, higher security, and same content can be sent to both website and mobile app via API.

Quick MCQs:

1. CSS stands for: Cascading Style Sheets | 2. Link tag: <a> | 3. Client-side language: JavaScript | 4. Not an image format: .mp3 | 5. Alt attribute: Alternative text for accessibility | 6. Include CSS: All of the above | 7. Line spacing: line-height | 8. Float property: Moves element left or right | 9. z-index: Stacking order (which sits on top) | 10. Table row tag: <tr>

11. Can WordPress rank on Google's first page?

👉 Ans: Yes, by optimizing content, keywords, site speed, and using SEO plugins like Yoast.

12. WordPress.com vs WordPress.org?

👉 Ans: WordPress.com = Hosted platform (limited control). WordPress.org = Self-hosted free software (full control).

13. Why use Jumbotron in Bootstrap?

👉 Ans: To make a big showcase banner for important headlines or offers.

Ex: `<div class="jumbotron"><h1>Welcome!</h1></div>`

14. Bootstrap Grid System?

👉 Ans: A mobile-first 12-column layout system using rows and columns (col-md-6).

15. Pseudo-class vs Pseudo-element?

👉 Ans: Pseudo-class (:hover) styles a state. Pseudo-element (::before, ::first-letter) styles a part of text.

16. What is Hoisting in JavaScript?

👉 Ans: JavaScript moves variable and function declarations to the top before running code.

Ex: `console.log(x); var x = 10; // Outputs undefined without error`

Quick MCQs:

1. Ternary operator: ?: | 2. Negotiation follow-up: Agreed by all parties | 3. IT risk action: Securing loose carpet tile to avoid tripping | 4. Color theory: Makes design appealing | 5. Visual hierarchy: Shapes and imagery | 6. Responsive grids: Adapt content to screens | 7. Trust color: Blue | 8. <iframe>: Embeds other web pages | 9. Prevent background repeat: background-repeat: no-repeat; | 10. Box shadow: box-shadow: 2px 3px 5px #888;

11. Why proper lighting in IT workspace?

👉 Ans: Reduces eye strain and headache, helping focus on screen.

12. Secondary colors in RGB?

👉 Ans: Cyan, Magenta, Yellow (CMY).

13. Smooth color change on hover:

```
button { background: blue; transition: background 0.3s ease; } button:hover { background: red; }
```

1. What is debugging?

👉 Ans: Finding and fixing bugs/errors in code.

2. WordPress Post vs Page?

👉 Ans: Post = Blog news with date and category. Page = Static information (About Us, Contact).

3. Why box-sizing: border-box?

👉 Ans: Keeps padding and border inside width so layout doesn't break.

4. DOM vs BOM?

👉 Ans: DOM = HTML page content (document.title). BOM = Browser window (window.location).

5. Block vs Inline elements?

👉 Ans: Block (<div>, <p>) takes full line width. Inline (, <a>) takes only content width.

6. JSON Array of 4 students:

```
let students = [ { "name": "Arif", "email": "arif@gmail.com", "mobile": "01313363836", "address":  
"Dhaka" }, { "name": "Rahim", "email": "rahim@gmail.com", "mobile": "01711223344", "address": "Dhaka"  
}, { "name": "Karim", "email": "karim@gmail.com", "mobile": "01811556677", "address": "Dhaka" }, {  
"name": "Jamal", "email": "jamal@gmail.com", "mobile": "01911889900", "address": "Dhaka" } ];
```

1. setTimeout() vs setInterval()?

👉 Ans: setTimeout runs ONCE after delay. setInterval runs REPEATEDLY every interval.

Ex: `setTimeout(() => alert("Hi"), 3000); | setInterval(updateClock, 1000);`

2. AJAX in jQuery?

👉 Ans: Loads data from server in the background without refreshing the web page.

Ex: `$.ajax({ url: "data.json", success: function(res) { ... } });`

3. Spell check in MS Word?

👉 Ans: Review tab → Spelling & Grammar, or press F7 shortcut key.

4. 4 Steps of Image Processing?

👉 Ans: 1) Acquisition (Capture), 2) Enhancement (Clean), 3) Segmentation (Cut), 4) Recognition (Identify).

5. nth-child() selector in CSS?

👉 Ans: Selects specific numbered children elements.

Ex: `tr:nth-child(even) { background: #eee; }` gives zebra lines to tables.

6. Default positioning in CSS?

👉 Ans: static (Normal document flow).

7. Text shadow in CSS:

Ex: `h1 { text-shadow: 2px 2px 4px gray; }`

8. Basic SVG tags:

👉 Ans: `<svg>`, `<circle>`, `<rect>`, `<line>`, `<path>`.

Job Sheet 1: Curriculum Vitae (CV)

👉 Ans: Professional CV with Photo, Summary, Education Table, Skills, and Contact Info.

রেডি ফাইল: d:\Web Design\cv.html এবং Desktop\CV\index.html

Job Sheet 2: Restaurant Menu / Booking Form

👉 Ans: Food items in cards/table with price, and a simple table reservation form.

Job Sheet 3: Responsive Bootstrap Layout

👉 Ans: Responsive Navbar, Hero Banner, and 3-column Service cards (col-md-4).

Job Sheet 4: Software Company Website (6 Pages)

👉 Ans: Complete 6-page responsive site already created in d:\Web Design\:

1) index.html (Home) | 2) about.html (About) | 3) services.html (Services)

4) projects.html (Portfolio) | 5) project-detail.html (Details) | 6) contact.html (Form)